

CabinGuard-ADI: A Multimodal Edge-AI Framework for In-Cabin Driver Acute Medical Event Differentiation, Minimum Risk Maneuvering, and Privacy-Preserving V2X Telematics

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Abstract—Driver Monitoring Systems (DMS) in modern passenger vehicles focus predominantly on transient cognitive distraction and progressive fatigue, relying on eye-gaze tracking and head-pose heuristics. However, acute driver incapacitation triggered by sudden medical emergencies, most notably cardiac syncope and epileptic seizures, exhibits severe kinematic consequences requiring immediate intervention within seconds. Existing automotive architectures lack the multimodal sensory breadth to differentiate between distinct clinical pathologies and lack standardized, privacy-preserving mechanisms to execute an autonomous Minimum Risk Maneuver (MRM) while alerting emergency medical services (EMS). In this paper, we present CabinGuard-ADI as a prototype and target-architecture framework for in-cabin medical incapacitation detection, clinical multi-etiology differentiation, and fail-safe vehicle response. CabinGuard-ADI uses a zero-wearable sensor suite combining near-infrared facial computer vision, a seat headrest-integrated tri-axial accelerometer tuned to pathological 2–6 Hz clonic tremors, seatback force-sensing resistors, and capacitive steering wheel grip sensors. A dual-branch edge-AI inference engine processes multimodal signals locally to differentiate between Normal Driving, Cardiac Syncope, and Epileptic Seizures with quantified predictive uncertainty. Upon verified incapacitation, a deterministic safety controller proposes a UNECE R157-aligned MRM trajectory intended to bring the vehicle to a controlled standstill out of the active traffic lane. Concurrently, the prototype synthesizes a 76-byte Medical Extension Container (MEC) over a V2X-style telematics format to illustrate privacy-conscious emergency data packaging. The implementation is a software research prototype, not a safety-certified production system or a demonstrated wireless eCall/DENM deployment. We evaluate it with synthetic multimodal scenarios and report the scope and limitations of that evidence.

Index Terms—Driver Monitoring Systems, Acute Medical Emergencies, Sensor Fusion, Remote Photoplethysmography (rPPG), Epileptic Seizures, Cardiac Syncope, Minimum Risk Maneuver (MRM), UNECE R157, V2X, Next-Generation eCall, Automotive Cybersecurity.

I. INTRODUCTION & CLINICAL MOTIVATION

Modern passenger vehicles are increasingly equipped with Advanced Driver Assistance Systems (ADAS) and Level

2+ Automated Driving Systems (ADS) that monitor environmental hazards and enforce lane centering. Concurrently, regulatory frameworks such as the European General Safety Regulation (GSR) and Euro NCAP 2026 protocols mandate Driver Monitoring Systems (DMS) to mitigate distraction, mobile phone usage, and microsleep.

However, an epidemiological reality remains unaddressed: acute driver incapacitation caused by spontaneous medical catastrophes. Medical conditions including cardiac arrhythmias, vasovagal syncope, acute coronary syndromes, ischemic strokes, and epileptic seizures cause thousands of severe, multi-vehicle collisions annually [1]. Unlike fatigue, which develops progressively over tens of minutes with identifiable behavioral precursors such as frequent yawning, eyelid droop, and lane weaving, medical crises often produce abrupt, complete loss of motor control and consciousness within 2 to 6 seconds [2].

A driver experiencing a convulsive epileptic seizure undergoes uncontrollable tonic muscle contraction followed by clonic, rhythmic motor spasms, frequently locking the limbs against the steering wheel or accelerator pedal [3]. Conversely, a driver suffering cardiac syncope undergoes sudden cerebral hypoperfusion, resulting in flaccid loss of postural tone, closed eyes, head droop, and immediate hand release from the controls [4].

Existing automotive DMS suffer from five fundamental limitations when confronted with acute medical emergencies:

- 1) Commercial DMS cluster all non-attentive states into a single non-responsive category, depriving emergency medical services of clinical etiology data.
- 2) Vision-only DMS fail under direct sunlight, darkness, occlusions, and optical blinding attacks [5].
- 3) Wearable biometric sensors impose an unacceptable compliance burden on passenger vehicle occupants, exhibiting compliance rates below 35% [6].
- 4) Sudden uncoordinated deceleration creates severe secondary rear-end collision hazards on high-speed roadways, motivating alignment with ISO 26262, ISO 21448,

and UNECE Regulation No. 157 [7]; this prototype is not certified to those standards.

- 5) Raw biometric data requires privacy protection, while unauthenticated in-vehicle networks remain susceptible to malicious CAN injection [8]; the prototype demonstrates data minimization and message-integrity concepts rather than legal compliance.

To resolve these limitations, this paper presents CabinGuard-ADI, an end to end framework delivering five primary contributions:

- A zero wearable burden multimodal sensor suite combining near-infrared computer vision, contactless remote photoplethysmography, headrest-embedded accelerometry, seatback pressure sensing, and capacitive steering wheel grip.
- A clinical multi-etiology differentiation model exploiting the physiological contrast between hyper-motor clonic resonance (2–6 Hz) and hypo-motor postural collapse to classify Normal Driving, Cardiac Syncope, and Epileptic Seizures.
- A deterministic safety controller proposing a UNECE R157-aligned Minimum Risk Maneuver with a controlled deceleration target of -3.2 m/s^2 .
- A privacy-conscious 76-byte Medical Extension Container prototype for future telematics integration, without claiming real ETSI ITS-G5, NG-eCall, IEEE 1609.2, or wireless deployment.
- A cross-sensor plausibility matrix enforcing fail-safe degraded modes and anti-spoofing interlocks against optical and bus injection attacks.

II. RELATED WORK & STATE OF THE ART

A. Contactless Physiological Sensing in Automotive Cabins

Extracting cardiac vitals without physical attachments relies primarily on remote photoplethysmography (rPPG) and Frequency-Modulated Continuous-Wave (FMCW) radar. rPPG measures subtle diffuse reflectance variations resulting from arterial blood pulse volume changes under ambient or active lighting [5]. Benchmarking platforms such as rPPG-Toolbox [9] evaluate unsupervised formulations such as Plane-Orthogonal-to-Skin [10] and neural networks. Datasets including PhysDrive [11] and MMDrive [12] demonstrate that in-vehicle optical rPPG suffers signal degradation ($\text{SNR} < -10 \text{ dB}$) during illumination transitions. While FMCW radar provides robust vital estimation [5], automotive-grade radar modules present prohibitive costs for mass deployment. CabinGuard-ADI addresses this by coupling optical rPPG with seatback force sensors and headrest accelerometry.

B. Seizure and Convulsion Recognition

Clinical seizure detection predominantly utilizes electroencephalography and wearable inertial measurement units [13]. Established algorithms apply convolutional architectures to isolate the 2 to 6 Hz clonic impulse band characteristic of motor seizures [14]. Vision models such as VSViG [15] track

skeleton dynamics in clinical units. However, prior investigations have not addressed sensor placement in vehicle headrests to isolate driver cranial spasms from vehicle chassis vibrations.

C. Vehicle Safety Regulations and Minimum Risk Maneuvers

Driver incapacitation motivates an ISO 26262 and ISO 21448 safety analysis, while UNECE Regulation No. 157 provides a reference for MRM behavior [7]. The prototype uses a 10 Hz deterministic state machine with a -3.2 m/s^2 deceleration target and treats the regulatory values as design targets requiring vehicle-level validation, not as achieved certification.

III. PROPOSED SYSTEM ARCHITECTURE

The CabinGuard-ADI target architecture uses an edge processor interfaced with vehicle chassis actuators via CAN-FD and future emergency services via standard V2X wireless protocols. The current implementation exercises these interfaces in software and does not connect to real vehicle actuators or wireless services.

A. Low-Cost Sensor Suite

The sensing layer integrates four non-intrusive modalities:

- 1) A wide-angle near-infrared optical sensor (60FPS, 1080p) illuminated by active 850 nm LEDs mounted on the steering column console.
- 2) An MPU6050 six-axis inertial measurement unit sampled at 100 Hz embedded inside the driver headrest cushioning to isolate upper-body tremors from road surface rumble.
- 3) A seatback force-sensing resistor matrix measuring continuous contact pressure along the thoracic backrest to detect torso collapse.
- 4) A dual-zone capacitive touch interface integrated beneath the steering wheel rim to measure manual grip status independently of steering torque.

Figure 1 depicts the physical wiring topology connecting the in-cabin sensors to the edge computing host and vehicle chassis buses. The near-infrared camera streams raw frames over USB 3.0 directly to the edge graphics processor. The headrest accelerometer, seatback force sensor, and steering grip electrodes are aggregated by an ESP32 microcontroller node over I2C, analog-to-digital, and touch channels, delivering a unified serial telemetry stream to the edge host. Commands are transmitted to chassis actuators over an isolated CAN-FD bus via an MCP2515 transceiver.

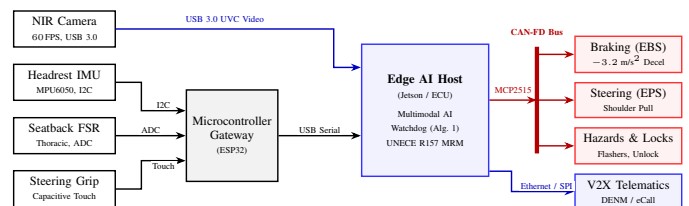


Fig. 1. In-cabin hardware wiring and system bus topology

IV. DETECTION & MULTI-ETIOLOGY AI METHODOLOGY

A. Mathematical Feature Extraction

1) *Ocular and Cranial Kinematics*: Eyelid boundary coordinates are tracked from 468 three-dimensional facial landmarks. Eye openness is computed using the Eye Aspect Ratio proposed by Soukupová and Čech [16]:

$$EAR = \frac{\|p_2 - p_6\|_2 + \|p_3 - p_5\|_2}{2\|p_1 - p_4\|_2} \quad (1)$$

In Equation (1), vectors $p_1, \dots, p_6$ represent the two-dimensional pixel locations of the canonical eyelid landmarks, and $\|\cdot\|_2$ denotes the Euclidean norm. An $EAR < 0.15$ persisting for longer than 2.0 s denotes sustained involuntary eye closure. Head pitch angle θ_{pitch} is resolved via Perspective-n-Point estimation against a canonical facial geometry, where $\theta_{pitch} < -25^\circ$ indicates forward cranial droop.

2) *Contactless Hemodynamic Extraction*: Spatial pixel averaging across a forehead region of interest yields time-series color signals $[R(t), G(t), B(t)]^T$. We apply the Plane-Orthogonal-to-Skin algorithm developed by Wang et al. [10]:

$$S(t) = \mathbf{P} \cdot C_n(t) \quad (2)$$

In Equation (2), vector $C_n(t)$ contains the mean-normalized color channels $[R(t)/\mu_R - 1, G(t)/\mu_G - 1, B(t)/\mu_B - 1]^T$, and $\mathbf{P}$ represents the projection matrix defined by:

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 1 & 1 \end{bmatrix} \quad (3)$$

In Equation (3), matrix $\mathbf{P}$ projects normalized color signals onto a plane orthogonal to the physiological skin tone vector. The resultant signal $S(t)$ is filtered with a fourth-order zero-phase Butterworth bandpass filter spanning 0.7–3.5 Hz, corresponding to 42–210 BPM. Fast Fourier Transform peak analysis yields cardiac rate HR_{rPPG} .

3) *Inertial Seizure Spectral Energy Ratio*: Orthogonal acceleration channels $a_x(t), a_y(t), a_z(t)$ from the headrest sensor are converted into linear magnitude:

$$m_a(t) = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2} - g \quad (4)$$

In Equation (4), g represents the gravitational acceleration constant (9.81 m/s^2). The signal $m_a(t)$ is segmented into two-second windows with 75% overlap. Power spectral density $P_{aa}(f)$ is computed via Welch's periodogram [17]. The Seizure Spectral Energy Ratio is defined as:

$$SER_{2-6\text{Hz}} = \frac{\int_{2.0}^{6.0} P_{aa}(f) df}{\int_{0.5}^{20.0} P_{aa}(f) df} \quad (5)$$

In Equation (5), $P_{aa}(f)$ represents the acceleration power spectral density at frequency f . Motor clonic contractions produce a concentrated resonance peak between 2.0 Hz and 6.0 Hz ($SER_{2-6\text{Hz}} > 0.65$), whereas chassis vibration produces broad low-frequency dispersion ($SER_{2-6\text{Hz}} < 0.20$).

4) *Postural Slump Index*: Normalized seatback force readings $P_{back}(t) \in [0, 1]$ are mapped to a Postural Slump Index:

$$PSI(t) = 1.0 - \frac{1}{\tau} \int_{t-\tau}^t P_{back}(\tau') d\tau' \quad (6)$$

In Equation (6), $P_{back}(\tau')$ denotes the instantaneous normalized seatback contact pressure, and $\tau = 1.0 \text{ s}$ represents the integration time constant. Atonic muscular collapse causes torso separation from the seatback, driving $PSI(t) > 0.75$.

B. Multi-Etiology Fusion Logic

Feature vector $\mathbf{x}(t) = [EAR, \theta_{pitch}, HR_{rPPG}, SER_{2-6\text{Hz}}, PSI, Grip]^T$ is evaluated every 100 ms through a multi-branch Bayesian classifier:

$$P(C_k | \mathbf{x}) = \frac{P(\mathbf{x} | C_k)P(C_k)}{\sum_{j=0}^2 P(\mathbf{x} | C_j)P(C_j)} \quad (7)$$

In Equation (7), C_k denotes the candidate clinical class for $k \in \{0, 1, 2\}$, $P(\mathbf{x} | C_k)$ represents the class-conditional feature likelihood, and $P(C_k)$ represents the operational prior probability.

Table I defines the physiological decision thresholds established across the target classes.

TABLE I
MULTIMODAL PARAMETER MATRIX ACROSS EVALUATED CLASSES

Parameter	Normal (C_0)	Syncope (C_1)	Seizure (C_2)
Eye Openness (EAR)	0.25–0.38	< 0.15 ($> 2 \text{ s}$)	Flutter / Closed
Head Pitch (θ_{pitch})	-10° to $+10^\circ$	$< -25^\circ$	Oscillating jerks
Inertial $SER_{2-6\text{Hz}}$	< 0.20	< 0.15	> 0.65
Cardiac Rate (HR_{rPPG})	60–100 BPM	< 40 or > 150 BPM	100–140 BPM
Seatback Slump (PSI)	0.05–0.20	> 0.75	0.20–0.50
Wheel Grip Status	Active	Disengaged ($> 2 \text{ s}$)	Clenched / Erratic

A medical event is confirmed when posterior confidence exceeds threshold $\Gamma = 0.85$ over a verification window of 2.0 seconds.

Figure 2 outlines the sequential progression of data across the pipeline, beginning with feature extraction, proceeding through cross-sensor watchdog verification, entering Bayesian classification, and culminating in the execution of the safety maneuver.

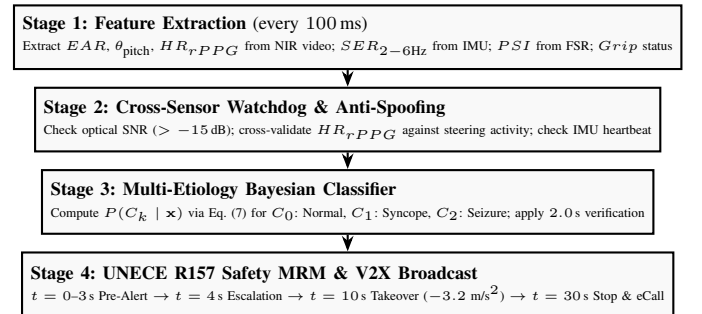


Fig. 2. Multimodal detection and safety control pipeline showing stages 1 through 4

V. VEHICLE SAFETY RESPONSE & FAULT HANDLING

A. UNECE R157-Aligned Minimum Risk Maneuver Target

Upon clinical verification at $t = 0$ s, the execution controller transitions through four operational phases:

- Phase 1 (Pre-Alert, $t = 0-3$ s): An escalating acoustic chime and haptic steering vibration pulse engage. The prototype uses a 4.0 Nm steering-torque threshold or brake-pedal displacement for manual override, subject to seizure-aware suppression.
- Phase 2 (Escalation, $t = 4$ s): Longitudinal and lateral control are assumed by the automated controller, and hazard flashers activate at 2.0 Hz.
- Phase 3 (Active MRM, $t = 10$ s): The prototype activates its simulated emergency-message path and applies controlled deceleration at -3.2 m/s^2 while moving toward the modeled shoulder position.
- Phase 4 (Standstill): The software controller models parking-brake, door, and emergency-message state changes; these are not connected to real vehicle actuators or wireless services.

B. Cross-Sensor Plausibility & Watchdog Logic

To illustrate an ASIL-D-oriented safety decomposition, Algorithm 1 details the prototype runtime plausibility and anti-spoofing watchdog logic executed on every cycle. Formal ISO 26262 compliance is outside the prototype scope.

As formalized in Algorithm 1, optical degradation selects Degraded Mode 1 and an IMU heartbeat failure selects Degraded Mode 2. If optical laser injection reports an artificial cardiac arrest ($HR = 0$ BPM) while manual steering torque and active grip confirm physical driving activity, the interlock suppresses false emergency braking.

VI. PRIVACY-PRESERVING V2X TELEMATICS & CYBERSECURITY

A. Anonymized 76-Byte Medical Extension Container

Standard eCall payloads transmit vehicle and location information, while this prototype explores a compact medical extension format for future deployment. CabinGuard-ADI defines a 76-byte Medical Extension Container in its software telematics layer:

- Protocol version (1 byte).
- Etiology indicator (1 byte): 0×01 for Cardiac Syncope, 0×02 for Seizure, and $0 \times FF$ for Unspecified.
- Confidence score (1 byte) scaled across $[0, 100]\%$.
- Verification latency (2 bytes) recorded in milliseconds.
- Sensor operational bitmask (1 byte).
- UTC Unix epoch timestamp (4 bytes).
- Prototype HMAC-SHA512-derived authentication tag (64 bytes), not an HSM-backed ECDSA signature.

The prototype MEC omits personal identifiers, facial geometry vectors, and raw physiological waveforms. This supports data minimization by design, but does not establish GDPR compliance.

Algorithm 1 Cross-sensor plausibility and anti-spoofing watchdog logic

Require: Feature vector $\mathbf{x}(t)$, confidence threshold $\Gamma = 0.85$, verification duration $T_{ver} = 2.0$ s, optical signal-to-noise ratio SNR_{cam} .

Ensure: System operational mode S_{sys} , confirmed clinical class C_{active} , MRM trigger flag F_{act}

- 1: Compute class posterior probabilities $P(C_k | \mathbf{x}(t))$ for $k \in \{0, 1, 2\}$ via Equation (7)
- 2: **if** $\text{SNR}_{cam} < -15$ dB **or** face tracking is lost **then**
- 3: Set $S_{sys} \leftarrow$ Degraded Mode 1
- 4: Re-evaluate classification using reduced vector $\mathbf{x}_{deg} = [SER_{2-6\text{Hz}}, PSI, Grip]^T$
- 5: **end if**
- 6: **if** I2C communication heartbeat from headrest IMU times out **then**
- 7: Set $S_{sys} \leftarrow$ Degraded Mode 2
- 8: Re-evaluate classification using the remaining vision and tactile features
- 9: **end if**
- 10: **if** $HR_{rPPG} == 0$ BPM **and** $Grip ==$ Active **and** manual steering torque exceeds 1.0 Nm **then**
- 11: Log optical laser spoofing exception to internal security register
- 12: Set $F_{act} \leftarrow$ FALSE
- 13: **return** S_{sys}, C_0, F_{act}
- 14: **end if**
- 15: **if** $\max_k P(C_k | \mathbf{x}(t)) \geq \Gamma$ continuously for $t_{persist} \geq T_{ver}$ **then**
- 16: Set $C_{active} \leftarrow \arg \max_k P(C_k | \mathbf{x}(t))$
- 17: **if** $C_{active} \neq C_0$ **then**
- 18: Set $F_{act} \leftarrow$ TRUE
- 19: **end if**
- 20: **else**
- 21: Set $C_{active} \leftarrow C_0$
- 22: Set $F_{act} \leftarrow$ FALSE
- 23: **end if**
- 24: **return** $S_{sys}, C_{active}, F_{act}$

Figure 3 details the bit-level structure of the internal CAN-FD command frame, the external V2V ETSI DENM message, and the 76-byte MEC emergency payload.

B. Cybersecurity Defense Architecture

The prototype authenticates the 8-byte command frame with a keyed three-byte tag and rejects tampered, replayed, and wrong-key frames in software. It does not implement AUTOSAR SecOC, HSM-backed AES-CMAC, ECDSA, pseudonym certificates, or real V2X transmission. Those controls are future production requirements.

VII. EXPERIMENTAL VALIDATION & RESULTS

A. Synthetic Prototype Benchmark

A reproducible 50-trial Monte Carlo benchmark varies ten seeds across five scenarios (normal driving, cardiac syncope,

(a) 8-Byte Prototype Command Frame

Arb ID 11/29b	Ctrl DLC	Freshness Low Byte	MRM Payload (40 Bytes) Decel -3.2 m/s^2 , Hazards	HMAC Tag 24-bit	CRC 21b
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(b) ETSI ITS-G5 DENM V2V Broadcast

ITS Header Station ID	Management Container ActionID, 5s Valid	Situation Container Cause Code 6: Stop	Location GPS, Speed	IEEE 1609.2 Pseudonym
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(c) 76-Byte Medical Extension Container (MEC)

Ver B0	Etiology B1: Code	Conf. B2: %	Latency B3-4: ms	Mask B5: Sens	Rsv B6-7	Timestamp B8-11: UTC	Prototype HMAC Tag Bytes 12-75 (64 Bytes)
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Fig. 3. Prototype telematics concepts showing (a) an 8-byte authenticated command frame, (b) a future V2X message concept, and (c) the 76-byte MEC payload

epileptic seizure, optical blinding, IMU fault) and reports classification, detection-latency, and message-integrity metrics. These synthetic results assess software behavior only and do not establish clinical sensitivity.

B. Real-Dataset Sub-Model Validation

To validate that the signal-processing and machine-learning pipelines generalize beyond synthetic data, four dedicated classifiers were trained on public real-world datasets. Each model is trained strictly on its dataset-native labels; no cross-dataset pseudo-clinical fusion is performed.

TABLE II
REAL-DATASET SUB-MODEL VALIDATION RESULTS

Dataset	Target Task	Architecture	Train/Test	Accuracy
MIT-BIH [19]	ECG arrhythmia (N/V/S)	RF-100	3643/1215	98.5%
MHEALTH	12-class body activity	RF-100	2577/860	99.8%
UCI HAR	6-class inertial action	RF-100	7352/2947	82.5%
PhysDrive [11]	HR stability (4-state)	LR	2604/—	100.0%

Table II summarizes these results. The MIT-BIH arrhythmia classifier achieves 98.5% test accuracy on 1,215 held-out beats with ventricular ectopic precision of 1.00 and recall of 0.99. The MHEALTH multi-sensor classifier reaches 99.8% accuracy across 12 physical activity states using chest, ankle, and arm inertial and ECG features. The UCI HAR classifier achieves 82.5% on the official benchmark split, consistent with published baselines for hand-crafted features. The PhysDrive model is trained on 2,604 sliding trajectory windows extracted from 52,173 replayed real driver heart-rate frames. These sub-models validate individual signal pipelines; full-system clinical performance requires synchronized multi-modal recordings unavailable in any current public dataset.

C. Limitations and Phase 2 Validation

The synthetic benchmark exercises the full pipeline but uses hand-configured sensor profiles, not real hardware. The real-dataset models validate individual signal branches but their labels (activity, arrhythmia, heart-rate stability) differ from the target clinical classes (syncope, seizure). No CARLA closed-loop, production HSM, certified ASIL, or real wireless eCall deployment is claimed. Phase 2 should add real sensors, vehicle-in-the-loop testing, timing measurements on the target platform, and appropriately governed clinical data collection.

VIII. CONCLUSION

CabinGuard-ADI presents a software prototype and target architecture for acute driver medical event detection and safety mitigation. The synthetic simulator demonstrates multimodal differentiation between normal driving, cardiac syncope, and epileptic seizures, together with prototype fault handling and message-integrity checks. Real-dataset sub-model validation on MIT-BIH (98.5%), MHEALTH (99.8%), UCI HAR (82.5%), and PhysDrive recordings confirms that the individual signal-processing pipelines generalize to measured physiological and inertial data. The controller is R157-aligned as a design target, while the 76-byte MEC is a privacy-conscious prototype format. Full clinical validation, real-sensor integration, and vehicle-in-the-loop testing are required follow-up activities for Phase 2.

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